Lecture 3

The Lebesgue integral: measure and convergence

Objectives. *Pointwise limits can leave the class of Riemann integrable functions. The Lebesgue integral allows more functions and gives convergence theorems with explicit hypotheses. We take measure and monotone convergence as standing facts. From them we prove Fatou's lemma, dominated convergence and a criterion for integrating an infinite sum.*

By the end of this lecture you will be able to:

- Identify a failure of completeness in the Riemann integral distance.
- Check the hypotheses of monotone and dominated convergence.
- Justify integration of a series by absolute integrability.
- Distinguish convergence almost everywhere from convergence in integral norm.

3.1 Where the Riemann integral fails

Integrals in this section are Riemann integrals. A uniform limit of Riemann integrable functions on a bounded interval is Riemann integrable, and the integrals converge by Theorem 1.2. Pointwise convergence does not have this property.

Enumerate the rational points of $(0,1)$ as $r_1, r_2, \dots$ and put $q_N = \mathbb{1}_{\{r_1,\dots,r_N\}}$ on $[0,1]$. Each function is Riemann integrable with integral zero: intervals of arbitrarily small total length cover the finite set where it is nonzero. Its pointwise limit is the indicator of the rational points in $(0,1)$. Every subinterval contains rational and irrational points, so this limit has every upper Darboux sum equal to 1 and every lower sum equal to 0. It is not Riemann integrable.

A second failure concerns the integral distance

$$\operatorname{dist}(f,g) = \int_0^1 |f(x) - g(x)| \, \mathrm{d}x.$$

Distinct functions can have distance zero. Even after identifying such functions, the Riemann integrable functions are not complete. To see this, surround each r_j by the interval

$$I_j = (r_j - 2^{-j-3}, r_j + 2^{-j-3}) \cap [0,1], \qquad G_N = \bigcup_{j=1}^{N} I_j, \qquad p_N = \mathbb{1}_{G_N}.$$

The interval lengths sum to at most $\sum_{j \geq 1} 2^{-j-2} = 1/4$. The functions p_N are Riemann integrable, and for $M > N$,

$$\operatorname{dist}(p_M, p_N) \leq \sum_{j>N} 2^{-j-2} = 2^{-N-2}.$$

Thus (p_N) is Cauchy in this distance.

Suppose a real Riemann integrable h were a limit. Since $\left|\int h - \int p_N\right| \le \operatorname{dist}(h, p_N)$, we would have $\int_0^1 h \le 1/4$.

Take any open subinterval of $(0,1)$. It contains some r_j and therefore a closed interval B of positive length inside I_j. For $N \ge j$, the function p_N is 1 on B, so $\int_B |h-1| \le \operatorname{dist}(h, p_N) \to 0$.

The supremum of h on B must be at least 1; otherwise this integral would be positive. Every subinterval therefore has supremum at least 1. All upper Darboux sums of h are at least 1, contradicting $\int_0^1 h \le 1/4$. There is no real Riemann integrable limit in this distance. A complex limit is also impossible: its real part would be a real limit, since $\operatorname{dist}(\operatorname{Re} h, p_N) \le \operatorname{dist}(h, p_N)$.

3.2 Standing facts for Lebesgue integration

We now use the Lebesgue integral. Its construction is supplied by the standing facts M1–M3. Under the course's proof policy, these cited facts do not change a subsequent proof's status to sketched.

Theorem 3.1 (M1: measure · *cited*). *For each $d \in \mathbb{N}$, there is a collection of* Lebesgue measurable *sets in $\mathbb{R}^d$, with a measure $|A| \in [0,\infty]$. It contains boxes and is closed under complements and countable unions. The measure of a box is the product of its side lengths. The measure is countably additive on pairwise disjoint measurable sets, and countably subadditive on arbitrary measurable unions. Every subset of a measurable set of measure zero is measurable and has measure zero.*

These properties are given in Stein–Shakarchi, *Real Analysis*, Chapter 1, §§2–3, including subsets of null sets in $\mathbb{R}^d$. The function-measurability properties below are in §4.1 of that chapter. A *null set* is a measurable set of measure zero. A property holds *almost everywhere*, abbreviated *a.e.*, if it fails only on a null set. Each point is null, so every countable set is null. In particular, the rational points form a null set even though they are dense. Our notation $|A|$ denotes measure here; for a number, $|z|$ denotes its modulus. Axler writes $\lambda(A)$ for Lebesgue measure of a measurable set.

Let $E \subset \mathbb{R}^d$ be measurable. A real or extended-real function f on E is *measurable* if $\{x \in E : f(x) > a\}$ is measurable for every $a \in \mathbb{R}$. A complex function is measurable if its real and imaginary parts are measurable. As part of M1, continuous functions are measurable, and measurability is preserved by the usual arithmetic operations where defined, countable suprema and infima of real or extended-real functions, pointwise limits, and composition with continuous functions where defined.

A nonnegative *simple function* has the form $s = \sum_{j=1}^N c_j \mathbb{1}_{E_j}$, where $c_j \in [0,\infty)$ and the measurable sets $E_j \subset E$ are pairwise disjoint. Its integral is

$$\int_E s = \sum_{j=1}^N c_j \left|E_j\right|.$$

We use $0 \cdot \infty = 0$ for nonnegative integration, including zero coefficients on sets of infinite measure. For a measurable $h : E \to [0,\infty]$, define

$$\int_E h = \sup\left\{\int_E s : s \text{ is simple and } 0 \le s \le h\right\}.$$

We also write $\int_E h(x)\,dx$ for this integral. It can be infinite. For measurable $A \subset E$, define $\int_A h = \int_E h\mathbb{1}_A$. The notation $\int_a^b$ means integration over $[a,b]$.

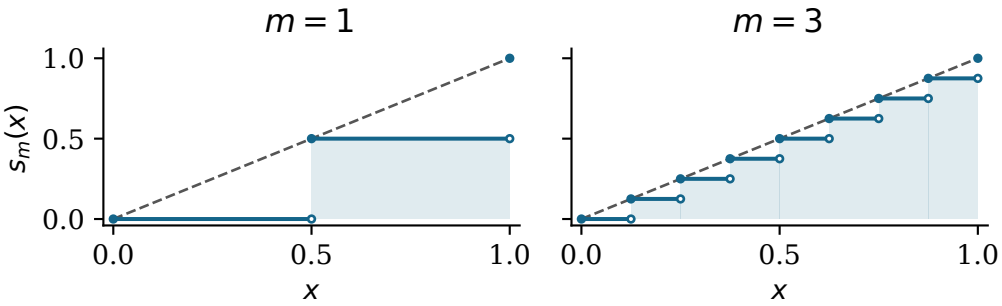
Theorem 3.2 (M2: nonnegative integrals · *cited*)**.** *The simple-function formula is independent of its representation, and agrees with the supremum definition on simple functions. The integral of nonnegative measurable functions is additive and homogeneous for nonnegative scalar factors, and preserves inequalities. Changing values on a null set does not change the integral. If $h_n : E \to [0,\infty]$ are measurable and $h_n(x)$ increases to $h(x)$ for every $x \in E$, then*

$$\int_E h = \lim_{n\to\infty} \int_E h_n. \tag{3.1}$$

This is the monotone convergence theorem.

For M2, see Axler, *Measure, Integration & Real Analysis*, §3A, including Theorems 3.7, 3.9 and 3.11. For example, $s = 2\mathbb{1}_{[0,1/2)} + 3\mathbb{1}_{[1/2,1]}$ has integral $2(1/2) + 3(1/2) = 5/2$. The indicator of a null set has integral zero, so the non-Riemann-integrable rational indicator from § 3.1 now has a well-defined integral equal to zero.

The dyadic lower steps give a concrete monotone approximation to $h(x) = x$ on $[0,1]$. Define $s_m(x) = 2^{-m}\lfloor 2^m x \rfloor$ for integers $m \geq 0$, where $\lfloor y \rfloor$ is the greatest integer at most y. The steps satisfy $s_m \leq s_{m+1} \leq x$ and $0 \leq x - s_m(x) < 2^{-m}$. The diagrams show the graphs for two values of m.



The horizontal segments show the lower steps, and the dashed line is $h(x) = x$. Filled and open markers specify the endpoint values.

The intervals have length 2^{-m}, so

$$\int_0^1 s_m(x)\,\mathrm{d}x = 2^{-2m} \sum_{j=0}^{2^m-1} j = \frac{1-2^{-m}}{2}.$$

The value at 1 does not affect the integral. By monotone convergence, $\int_0^1 x\,\mathrm{d}x = 1/2$.

A measurable $f : E \to \mathbb{C}$ is *integrable* if $\int_E |f| < \infty$. This is the meaning of integrable from now on. For real f, write $f^+ = \max(f, 0)$ and $f^- = \max(-f, 0)$, so $f = f^+ - f^-$. Define $\int_E f = \int_E f^+ - \int_E f^-$. For complex f, integrate its real and imaginary parts separately.

Theorem 3.3 (M3: agreement with calculus · *cited*)**.** *A Riemann integrable function on a compact interval is Lebesgue measurable, and its Lebesgue and Riemann integrals agree. Let J be an interval, possibly unbounded, and let f be Riemann integrable on every compact subinterval of J. If the improper integral of $|f|$ over J is finite, then the Lebesgue integral agrees with the improper Riemann integral of f.*

The compact-interval clause is Axler, §3B, Theorem 3.34. For the improper clause, exhaust J by increasing compact intervals J_m. The compact-interval agreement and MCT give $\int_J |f| = \lim_m \int_{J_m} |f| < \infty$. For real f, apply MCT to $f^+\mathbb{1}_{J_m}$ and $f^-\mathbb{1}_{J_m}$ and subtract their

finite limits. The complex case uses real and imaginary parts. These are the standing results we use from measure theory. In particular, Lebesgue integration retains the familiar values of convergent calculus integrals. Conditional improper integrals require a separate argument in Lecture 4.

3.3 Integrable functions and dominated convergence

The integral on integrable functions is linear. The inequality $|f+g| \le |f|+|g|$ first shows that $f+g$ is integrable. For real integrable f and g, the identity $(f+g)^+ + f^- + g^- = (f+g)^- + f^+ + g^+$ reduces additivity to M2. All terms have finite integral, so subtraction is legitimate. Scalar linearity follows from positive and negative parts. The complex extension is linear coordinate by coordinate. Monotonicity for real integrable functions follows by integrating their nonnegative difference. The integral also satisfies

$$\left| \int_E f \right| \le \int_E |f| . \tag{3.2}$$

Indeed, for a nonzero integral I, put $c = \bar{I}/|I|$. Then $|I| = \mathrm{Re}(cI) = \int_E \mathrm{Re}(cf) \le \int_E |f|$. The zero-integral case is immediate.

For nonnegative measurable h and $a > 0$, monotonicity gives $a\,|\{h > a\}| \le \int_E h$. If the integral is finite, the set where $h = \infty$ has measure zero, since its measure is at most $(\int_E h)/a$ for every $a > 0$. Also, $\int_E h = 0$ holds exactly when $h = 0$ a.e.: apply the same bound with $a = 1/j$ and take the countable union over $j \in \mathbb{N}$.

The quantity $\|f\|_1 = \int_E |f|$ is a *seminorm*: it has the norm properties except that it can vanish for a nonzero function. For example, $\mathbb{1}_{\{0\}}$ on $[0,1]$ is nonzero but has seminorm zero. Identify two functions when they agree a.e. The resulting equivalence classes form $L^1(E)$, and $\|\cdot\|_1$ is a norm on these classes. Convergence *in* L^1 means $\|f_n - f\|_1 \to 0$. Lecture 5 proves completeness of these spaces.

For a sequence (a_n) in the extended real numbers, its *lower limit* and *upper limit* are

$$\liminf_n a_n = \lim_{N\to\infty} \inf_{n\ge N} a_n, \qquad \limsup_n a_n = \lim_{N\to\infty} \sup_{n\ge N} a_n.$$

The tail infima increase and the tail suprema decrease, so these limits exist.

Lemma 3.4 (Fatou). *For nonnegative measurable $h_n : E \to [0,\infty]$,*

$$\int_E \liminf_{n\to\infty} h_n \le \liminf_{n\to\infty} \int_E h_n.$$

Proof. Put $v_N(x) = \inf_{n\ge N} h_n(x)$. These functions are measurable and increase to $\liminf_n h_n$. Since $v_N \le h_n$ for every $n \ge N$, $\int_E v_N \le \inf_{n\ge N} \int_E h_n$. Use monotone convergence on the left and pass to the increasing limits. ∎

For example, $h_n = \mathbb{1}_{[n,n+1)}$ on $\mathbb{R}$ tends pointwise to zero, but $\int_{\mathbb{R}} h_n = 1$. Fatou's inequality can therefore be strict.

Theorem 3.5 (Dominated convergence). *Let $f_n : E \to \mathbb{C}$ be measurable and converge a.e. to a measurable $f : E \to \mathbb{C}$. Suppose a nonnegative integrable g satisfies $|f_n| \le g$ a.e. for every n. Then f_n and f are integrable, and*

$$\|f_n - f\|_1 \longrightarrow 0, \qquad \int_E f_n \longrightarrow \int_E f.$$

Proof. The exceptional sets for convergence and the countably many bounds have a null union. Set all functions to zero there. The integral is unchanged. Now $|f| \le g$ everywhere, so f and all f_n are integrable. Put $a_n = \int_E |f_n - f|$. The functions $2g - |f_n - f|$ are nonnegative and converge to $2g$. Fatou gives

$$2\int_E g \le \liminf_n \left(2\int_E g - a_n\right) = 2\int_E g - \limsup_n a_n.$$

The number $2\int_E g$ is finite. Subtraction gives $0 \le \limsup_n a_n \le 0$, so $a_n \to 0$. Finally, Equation (3.2) bounds the difference of the integrals by a_n. ∎

A majorant must be independent of the sequence index. A separate finite integral for each term does not supply it. On a finite-measure set a common constant bound is integrable. On the whole line a positive constant bound is not integrable, so decay at infinity must also be controlled. These checks concern the integration set as well as the formula for the terms.

3.4 Limits and infinite sums under the integral

On $[0,1]$, the functions $f_n(x) = x^n$ tend to zero except at 1. They satisfy $|f_n| \le 1$, and this dominating function is integrable on the finite interval. Dominated convergence gives $\int_0^1 x^n \, dx \to 0$. By M3, the exact value is $1/(n+1)$, as in real calculus. The convergence is not uniform, since the supremum of x^n on $[0,1)$ is 1.

Theorem 3.6 (Absolute integration of a series)**.** *Let $f_n : E \to \mathbb{C}$ be integrable and suppose*

$$\sum_{n=1}^{\infty} \int_E |f_n| < \infty.$$

Then $\sum_n f_n$ converges absolutely a.e. to an integrable function f, and its partial sums converge to f in L^1. Moreover,

$$\int_E \sum_{n=1}^{\infty} f_n = \sum_{n=1}^{\infty} \int_E f_n. \tag{3.3}$$

On the exceptional null set the sum may be assigned the value zero.

Proof. By monotone convergence applied to $\sum_{n=1}^{N} |f_n|$, the function $g = \sum_n |f_n|$ has finite integral. It is finite a.e., so the series converges absolutely there. Set its sum to zero elsewhere; it is measurable by M1. After setting g to zero on that null set, it is an integrable dominator. For $S_N = \sum_{n=1}^{N} f_n$, we have $|S_N| \le g$ a.e. Dominated convergence proves the claim about the integrals and the L^1 limit. Directly, its error satisfies

$$\|f - S_N\|_1 \le \sum_{n>N} \int_E |f_n| \longrightarrow 0,$$

where monotone convergence also justifies the integral of the nonnegative tail. ∎

For example, take $f_n(x) = (-1)^{n-1} x^{n-1}/2^n$ on $[0,1]$. Here $\int_0^1 |f_n| = 1/(n2^n) \le 2^{-n}$, so the theorem applies. The geometric sum is $\sum_n f_n(x) = 1/(2+x)$. Its integral gives

$$\ln(3/2) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n2^n}.$$

The same integrable function must dominate the whole sequence. The moving bumps $f_n = n\mathbb{1}_{(0,1/n)}$ of § 1.1 tend pointwise to zero but have integral 1. Suppose an integrable $g \geq 0$ dominated every f_n a.e. On the disjoint intervals $A_m = (2^{-m-1}, 2^{-m})$, for $m \geq 1$, we would have $g \geq f_{2^m} = 2^m$ a.e. Hence

$$\int_{A_m} g \geq 2^m \, |A_m| = \frac{1}{2}.$$

Summing over any N of these intervals gives $\int_0^1 g \geq N/2$, a contradiction. The dominated convergence theorem cannot be applied here.

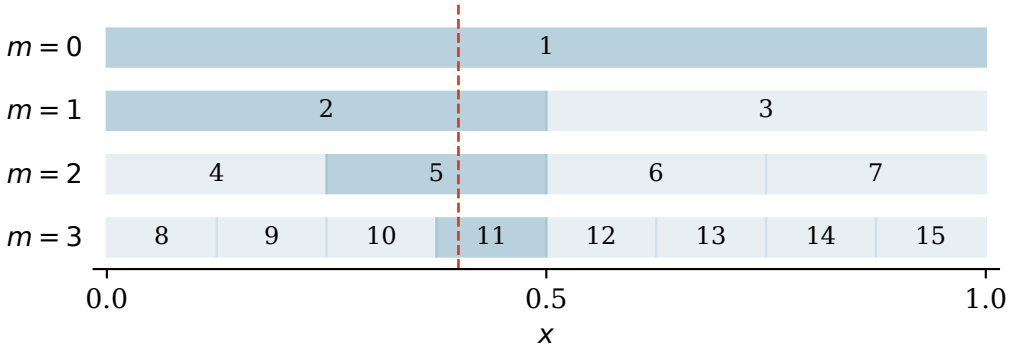
The new integral also supplies the missing limit from § 3.1. Let $G = \bigcup_{j \geq 1} I_j$. Monotone convergence gives $\int_0^1 \mathbb{1}_G = \lim_M \int_0^1 p_M \leq 1/4$. For fixed N, the functions $p_M - p_N$ increase to $\mathbb{1}_G - p_N$ as $M \to \infty$, with $M \geq N$. By M3, monotone convergence and the earlier distance bound, $\|\mathbb{1}_G - p_N\|_1 \leq 2^{-N-2}$. Thus this sequence converges in the Lebesgue integral norm, although no Riemann integrable function represents its limit. Such a representative would contradict the earlier no-limit argument by M3.

3.5 Convergence almost everywhere and in integral norm

Convergence in L^1 need not give convergence a.e. The *typewriter sequence* on $E = [0, 1)$ lists the indicators of successive dyadic intervals. For integers $m \geq 0$ and $0 \leq j < 2^m$, set

$$f_{2^m+j}(x) = \mathbb{1}_{[j2^{-m},(j+1)2^{-m})}(x), \qquad x \in E. \tag{3.4}$$

The indices in block m run from 2^m to $2^{m+1} - 1$. The diagram shows their intervals and indices.



Each box represents the half-open interval where one function equals 1, labelled by its index. The vertical line $x = 2/5$ meets one such interval in every block.

Every interval in block m has length 2^{-m}, so $\|f_{2^m+j}\|_1 = 2^{-m}$. The block number tends to infinity with the index, and therefore $f_n \to 0$ in L^1. For each $x \in [0, 1)$, exactly one interval in every block contains x. Thus $f_n(x) = 1$ infinitely often. Every block with $m \geq 1$ also has an interval missing x, so $f_n(x) = 0$ infinitely often. The sequence does not converge at any point of $[0, 1)$, including dyadic endpoints.

The reverse implication also fails: the moving bumps converge pointwise to zero, but their L^1 distance from zero is always 1. Under the domination hypothesis, convergence a.e. also gives convergence in L^1. Lecture 5 proves that an L^1-convergent sequence always has an almost-everywhere convergent subsequence.

Summary.

- M1–M3 supply measure, nonnegative integration, monotone convergence and agreement with absolutely convergent calculus integrals.
- Fatou bounds the integral of a lower limit for nonnegative measurable functions.
- For measurable f_n and f with $f_n \to f$ a.e., a nonnegative integrable g with $|f_n| \le g$ a.e. for every n gives convergence in L^1 and $\int f_n \to \int f$.
- A series of measurable complex functions with $\sum_n \int |f_n| < \infty$ converges absolutely a.e. and may be integrated term by term.
- Convergence a.e. and convergence in L^1 do not imply one another without further hypotheses.

Lecture 4 applies these convergence results to integrals with parameters.

References

S. Axler, *Measure, Integration & Real Analysis*, Springer GTM 282, 2020, online version 16 August 2025, §§2D, 3A–3B. Fatou is Exercise 17 in §3A. E. M. Stein and R. Shakarchi, *Real Analysis*, Princeton, 2005, Chapter 1, §§2–3. R. S. Strichartz, *The Way of Analysis*, revised edition, 2000, §§14.1 and 14.3.